

AlphaBench

Can a model predict the direction of daily equity returns well enough to beat buy-and-hold after costs?

Answer: no — and the value is in how carefully that was established.

Veerendra Kosuri · Walk-forward equity return forecasting & backtesting

Educational research artifact — not investment advice. All results are historical simulations with no guarantee of future performance.

1 · The problem, reframed

The naive version, which nearly every "stock prediction" project builds: predict tomorrow's *price*.

It fails for two reasons at once:

- Prices are non-stationary. Confirmed here: **0/29 symbols stationary** under both ADF and KPSS. Returns are stationary on both: **29/29**.
- A model trained on price levels wins on RMSE by predicting "roughly today's price." It scores beautifully and has learned nothing.

The version actually built: predict the *sign* of the forward return, on a volatility-scaled deadband — is the next move large enough, in either direction, to be worth acting on?

Labels: $y = 1$ if $\text{fwd_ret} > \kappa \cdot \sigma$, 0 if $\text{fwd_ret} < -\kappa \cdot \sigma$, dropped otherwise. The deadband stops the target being dominated by whichever volatility regime happens to prevail.

2 · Methodology — the parts that decide whether a result is real

Walk-forward, expanding window. Seven folds, validating 2017 → 2023. Never a random k-fold: shuffling a time series lets the model train on the future.

Purge + embargo. A 5-day purge drops training rows whose label window overlaps the validation block; a 5-day embargo follows it. Without these, the last rows of training leak the first days of validation.

Leakage caught structurally, not by inspection.

- Every feature is forced through one `safe_shift` before it reaches a model — a blanket rule that cannot be forgotten per-feature.
- The test suite trains on a synthetic random walk and *fails the build* if AUC departs from chance, or if any feature correlates > 0.10 with the forward return.
- Strongest observed correlation with tomorrow's return across ~50 features: **0.0196**.

The holdout is sealed and single-use. 2025-01-01 onward, never touched during development; the CLI refuses to score it twice.

3 · The baseline ladder — B0 first

A number without its baseline is not a result.

	Model	ROC-AUC	Accuracy
B0	Persistence (today's direction → tomorrow's)	0.487	0.488
B0	Majority class	0.500	0.518
B1	Logistic regression (5 features)	0.510	0.518
B2	ARIMA(1,0,1) on log returns	0.503	0.511
M1	LightGBM — primary, and the model served	0.506	0.518
M2	XGBoost	0.515	0.518
M3	LSTM (PyTorch, CPU, small)	0.515	0.518
M4	Rank-average ensemble (M1+M2+M3)	0.514	0.508

Every entry sits in **0.487 – 0.515**. Each model's own fold-to-fold standard deviation is **0.01 – 0.02**.

The gaps between models are smaller than the noise within any one model. There is no ladder to climb here.

4 • Two results reported against expectation

The LSTM did not lose. PROPOSAL §4.2 predicted it would underperform the GBM at this data scale. It came in at 0.515 — tied with XGBoost, marginally *above* LightGBM's 0.506.

Reported as it happened rather than adjusted to match the prediction. It also doesn't contradict the prediction in any way that matters: every confidence interval overlaps every other, so "M3 beat M1" is not a claim this sample size supports.

Prophet was rejected, not run. It decomposes a series into trend + seasonality + holiday components — structure daily equity returns demonstrably lack (the ACF is flat beyond lag 0). It would fit smooth curves to noise, convincingly, because that is what a smoothing model does regardless of whether structure is present. A model guaranteed to produce plausible-looking output on data it has no business modelling invites false confidence. B2 ARIMA fills the classical-baseline role on a specification that at least matches the data-generating process.

5 · Economics — where the result actually dies

Metric	Strategy	Buy-and-hold NIFTY
Sharpe (annualised, net of 10 bps)	0.108	1.799
Annual return	1.00%	34.96%
Max drawdown	−29.1%	−37.5%

Excess Sharpe: −1.69.

Cost sensitivity — the edge does not survive being traded:

Total cost	0 bps	2 bps	5 bps	10 bps	20 bps
Sharpe	0.353	0.303	0.230	0.108	−0.131

Even *gross* of costs the Sharpe is 0.35. It crosses zero between 10 and 20 bps — a realistic commission-plus-slippage assumption is enough to erase it entirely.

6 · How confident are we in "no"?

Block-bootstrap Sharpe (21-day blocks, 2000 resamples): 0.155, 95% CI [**-0.45, 1.13**], $P(\text{Sharpe} > 0) = 0.699$. The interval comfortably contains zero.

Deflated Sharpe ratio. 50 Optuna trials were run during development. Correcting the observed Sharpe for that search effort:

	Value
Observed Sharpe	0.108
Null bar under 50 trials	0.146
$P(\text{skill is real})$	0.46

The null bar is higher than the Sharpe actually achieved — the best of 50 random configurations with zero real skill would be expected to look *better* than this did.

7 • The sealed holdout — scored once

2025-01-01 onward, 5,128 rows, `models/lightgbm_h1/final.joblib`, run exactly once.

Metric	Value
ROC-AUC	0.5047
95% CI (block bootstrap)	[0.4817, 0.5266]
Backtest Sharpe, net	−0.583
Buy-and-hold Sharpe	0.726

The AUC interval **straddles 0.50**. The strategy is loss-making out of sample against a benchmark that rose.

This is the number the whole design existed to protect: nothing was tuned after seeing it, and it is reported unchanged.

8 · Does it replicate? A second market

The identical pipeline, same hyperparameters, no retuning, rerun on 30 US large-caps (benchmark SPY) in fully separate data and model paths.

	NSE (primary)	US (generalisation)
Walk-forward AUC	0.506	0.519
Backtest Sharpe, net	0.108	0.349
Buy-and-hold Sharpe	1.799	1.053
Excess Sharpe	−1.69	−0.70

Same qualitative finding in both: a small, statistically unreliable edge that does not survive costs against buy-and-hold.

That it replicates across two markets with different data-quality profiles and no shared tuning argues against "this NSE universe was unlucky" and for "this feature set, at this data scale, does not carry a robust directional signal."

9 · Interpretability — and what it says

SHAP on the final model: top features are **market-driven** (`mkt_ret_5d` , `ret_1d` , `mkt_ret_1d` , `mom_5d` , `mkt_vol_21d`) rather than stock-specific — whatever is being picked up leans toward broad market beta.

Feature stability across folds: mean pairwise Jaccard overlap of the top-10 = 0.352.

A feature that matters in one fold is roughly as likely to drop out of the top 10 next fold as to stay. That is evidence *against* a stable exploitable signal, not for one — a genuine edge should show more consistent importance through time.

The correlation matrix says the same thing from another angle: `ret_1d` and `mom_1d` correlate at **1.000** (the same quantity computed two ways), with several other pairs above 0.9. ~50 columns, far fewer independent things.

10 · Limitations — stated, not buried

Survivorship bias. The universe is today's constituents, back-tested through history — firms that delisted or fell out of the index are absent. Point-in-time constituent data isn't freely available, so this is inherited rather than fixed. It inflates the *benchmark* more than the direction-classification task, but is not zero.

Single-market primary result, mitigated but not eliminated by the US replication.

Daily bars only. No intraday microstructure, no order book, no fundamentals, no news or sentiment — all explicitly out of scope, all plausibly where remaining signal lives.

Costs are modelled, not measured. A flat 10 bps commission-plus-slippage; real execution has market impact that scales with size.

One flagged data row (−76% one-day move in NESTLEIND.NS, Jan 2010 — an unadjusted corporate action) is left in rather than hand-patched. It sits seven years before the first validation fold.

11 · What I'd do next

Accept the null result and change the question, rather than keep tuning this one. The deflated Sharpe already says the search effort spent exceeds the signal found; more search on the same features would produce a better-looking number and a worse-supported one.

Ordered by expected value:

1. **Lower the frequency.** Daily direction is close to a martingale difference sequence. Weekly or monthly horizons have a better signal-to-noise ratio and — decisively — proportionally far lower cost drag, which is what actually killed this result.
2. **Add genuinely orthogonal data.** Fundamentals, analyst revisions, news sentiment. ~50 technical features derived from OHLCV are close to one view of the same thing.
3. **Predict something other than direction.** Volatility *is* forecastable — the ACF of |returns| decays slowly where the ACF of returns dies at lag 1. A vol-targeting or risk-model application uses the structure that is demonstrably there.
4. **Cross-sectional ranking rather than per-name timing.** Relative ordering within a universe is a better-posed problem than absolute direction, and it is what most working equity strategies actually do.

What this project demonstrates

Not a profitable trading strategy. It says so, everywhere, in its own README.

A validation harness rigorous enough that a negative result can be trusted:

- Leakage prevented structurally and enforced by tests that fail the build
- Walk-forward with purge and embargo, never a random split
- A full baseline ladder, B0 first, with every model reported honestly — including the two that contradicted expectations
- Costs applied, swept, and shown to be decisive
- Distribution-free confidence intervals, and a deflated Sharpe that accounts for search
- A sealed holdout, scored once, reported unchanged
- Replicated on a second market with no retuning

Two bugs found by refusing to accept a good-looking number: an $h=5$ backtest reporting a Sharpe above 3 and a >700% single year (overlapping h -day return windows compounded as if daily), and — while fixing that — a second, subtler version where per-symbol downsampling failed to synchronise across symbols. Both are now covered by regression tests. A result that looks too good is treated as a defect until proven otherwise.

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